Unit Test Report — Sprint 2

Tester: Hanumantha L (PES2UG24CS179) **Application:** `railway_sprint2.cpp` **Source baseline:** `fc2c53227724ae2fd914a0d1c4fb4815c1ee2727` **Environment:** Linux x86-64, GCC/G++ C++17 **Evidence:** `docs/evidence/sprint2/`

Method

The implementation is a single 620-line CLI translation unit. It does not expose a unit-test target, headers, dependency injection boundary, or framework tests. Therefore, a separate Unity/CUnit unit suite could not be executed without modifying the application design, which was outside this QA task. Function behavior was exercised through deterministic CLI probes and file assertions instead; these are system-level black-box checks, not claimed as isolated unit tests.

Function-level behaviors exercised

Behavior	Input/probe	Actual result	Status
`validDate` through search	`2026-02-31`, then valid date	Invalid date rejected; valid date accepted	PASS
`readInt` bounds	Menu `99`, seat request `999`	Range error and recovery	PASS
Registration duplicate check	Existing contact `9000000001`	Duplicate rejected	PASS

Password hashing/storage	Register Alice; inspect `users.txt`	Hash field stored, not plaintext	PASS
Lockout state transition	Three wrong passwords	Account locked	PASS
Booking PNR generation	Successful booking	Numeric PNR created	PASS
Fare/payment mapping	UPI, Cash, Card bookings	Selected modes and fare displayed	PASS
Cancellation/ refund calculation	Cancel confirmed booking	10% charge and 90% refund displayed	PASS
Ownership guard	Bob cancels Alice's PNR	Rejected as not owned	PASS
Admin state operations	Add/update/ inventory/report/ remove	Each operation succeeded	PASS

Build command

```
g++ -std=c++17 -Wall -Wextra -pedantic --coverage railway
```

The build completed without compiler errors. The complete command transcripts and outputs are under `docs/evidence/sprint2/`.

Limitations

No isolated unit-test executable was created, no source was refactored to make functions linkable, and no unit framework was introduced. A future developer task should extract Auth, Search, Booking, Cancellation, and Admin modules behind testable interfaces and add

true unit tests.